

Since i don't have a public domain, I will use a **self-signed SSL/TLS certificate** for testing. This is perfect for labs, but browsers will show a warning because it's not issued by a trusted CA — which is fine here.

Task 2 — Step-by-Step Guide (Apache, Self-Signed SSL/TLS)

1 Install SSL/TLS Support for Apache

```
sudo apt update  
sudo apt install apache2 openssl ssl-cert  
sudo a2enmod ssl
```

2 Create a Self-Signed SSL Certificate

We'll store it in /etc/ssl and set it to be valid for **1 year** (365 days).

```
sudo mkdir -p /etc/ssl/private  
sudo openssl req -x509 -nodes -days 365 -newkey rsa:2048 \  
-keyout /etc/ssl/private/apache-selfsigned.key \  
-out /etc/ssl/certs/apache-selfsigned.crt
```

When prompted, enter:

- **Country Name:**
 - **State/Province:**
 - **Locality Name:**
 - **Organization Name:** CosmiCode
 - **Organizational Unit:** IT
 - **Common Name:**
 - **Email Address:**
-

3 Configure Apache for SSL

Create a new SSL-enabled virtual host file:

```
sudo nano /etc/apache2/sites-available/ssl-test.conf
```

Paste this configuration:

```
<VirtualHost *:443>
```

```
    ServerName localhost
```

```
    DocumentRoot /var/www/html
```

```
    SSLEngine on
```

```
    SSLCertificateFile /etc/ssl/certs/apache-selfsigned.crt
```

```
    SSLCertificateKeyFile /etc/ssl/private/apache-selfsigned.key
```

```
<Directory /var/www/html>
```

```
    Options Indexes FollowSymLinks
```

```
    AllowOverride All
```

```
    Require all granted
```

```
</Directory>
```

```
ErrorLog ${APACHE_LOG_DIR}/ssl_error.log
```

```
CustomLog ${APACHE_LOG_DIR}/ssl_access.log combined
```

```
</VirtualHost>
```

Save & exit (CTRL+O, ENTER, CTRL+X).

Enable SSL Site & Restart Apache

```
sudo a2ensite ssl-test.conf
```

```
sudo systemctl restart apache2
```

5 Configure Strong SSL/TLS Protocols

Edit SSL settings file:

```
sudo nano /etc/apache2/mods-available/ssl.conf
```

Find and update:

```
SSLProtocol all -SSLv2 -SSLv3 -TLSv1 -TLSv1.1
```

```
SSLCipherSuite HIGH:!aNULL:!MD5
```

```
SSLHonorCipherOrder on
```

Save & restart Apache:

```
sudo systemctl restart apache2
```

Verify SSL/TLS is Working

Local browser test:

Visit:

```
https://localhost
```

You'll see a warning about self-signed certificate — accept it to proceed.

Command-line test:

```
openssl s_client -connect localhost:443
```

You should see certificate details and **SSL handshake success**.

Protocol & Cipher Test:

```
nmap --script ssl-enum-ciphers -p 443 localhost
```

(This shows all supported protocols and ciphers.)

Test with a Simple HTTPS Request

```
curl -vk https://localhost
```

The -k option ignores the self-signed certificate warning.

If you follow these exactly, you'll have:

- A working **SSL/TLS-enabled Apache server**
- Secure protocol configuration (no SSLv2, SSLv3, TLS1.0, TLS1.1)
- Verified encryption with openssl and nmap